

Basic Principles of Medicine 1
Module: Cardiovascular, Pulmonary, Renal
Lecture: CPR 40

Development and Clinical Embryology of the Urinary system

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Objectives

SOM.MK.II.BPM1.3.CPR.2.ANAT.1464	Describe the development, location, blood supply, innervation and venous drainage of the suprarenal glands.
SOM.MK.II.BPM1.3.CPR.2.ANAT.1465	Describe the development of the kidney from the metanephric diverticulum and the metanephric mass of mesoderm.
SOM.MK.II.BPM1.3.CPR.2.ANAT.1466	Describe the development of the ureters.
SOM.MK.II.BPM1.3.CPR.2.ANAT.1467	Describe the development of the urinary bladder and urethra.
SOM.MK.II.BPM1.3.CPR.2.ANAT.1468	Describe the development of a discoid (pancake) and a horseshoe kidney.
SOM.MK.II.BPM1.3.CPR.2.ANAT.1469	Describe the development of a bifid or double ureter.
SOM.MK.II.BPM1.3.CPR.2.ANAT.1470	Describe the embryologic mechanism responsible for renal agenesis.
SOM.MK.II.BPM1.3.CPR.2.ANAT.1471	Describe how a kidney may have more than one renal artery and how an accessory renal artery may cause hydronephrosis.

Recommended reading

Gary C. Schoenwolf, PhD; Steven B. Bleyl, MD, PhD; Philip R. Brauer, PhD; Philippa H. Francis-West, PhD

[Larsen's Human Embryology](#), Ch15, 368-385 [Link](#)

T.V.N. (Vid) Persaud, MD, PhD, DSc, FRCPath (Lond.), FAAA; Mark G. Torchia, MSc, PhD

[The Developing Human](#), Ch 12, 225-264 [Link](#)

Pay special attention to the **Green/ blue** boxes for the clinical correlations for this topic.

Images taken from:

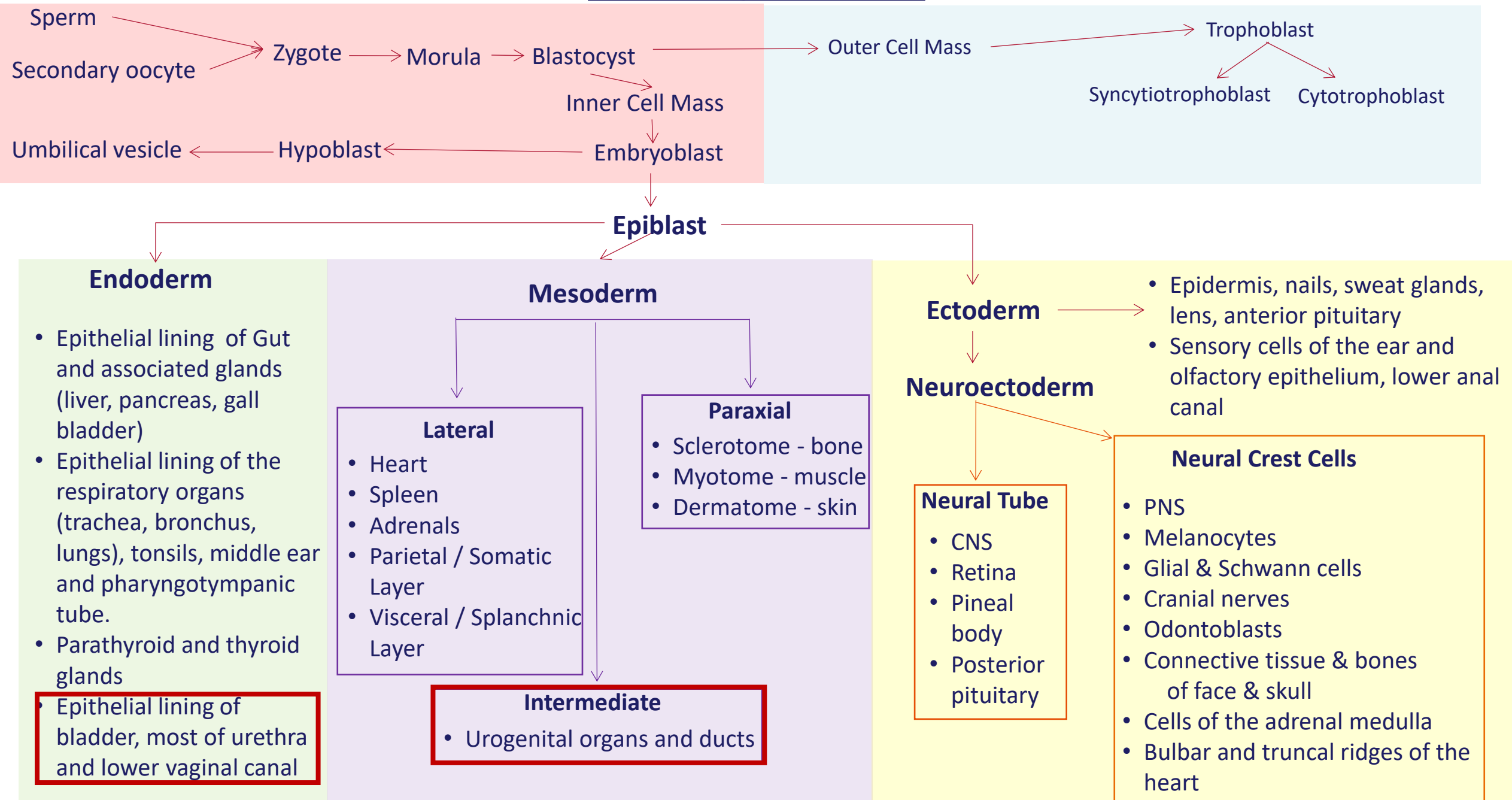
The Developing Human Clinically Oriented Embryology, 11th Edition. Keith L. Moore, T.V.N. Persaud, Mark G. Torchia. *Elsevier*

Langman's Medical Embryology, 14th Edition. T.W. Sadler. *Wolters Kluwer*

Peter H. Abrahams, Jonathan D. Spratt – McMinn's & Abrahams' Clinical Atlas of Human Anatomy, 7th Edition (2013, Mosby).

Atlas of Human Anatomy, 6th Edition – Frank H. Netter, M.D.

The Big picture



Terminology

- Allantois – a diverticulum consisting of endothelium and mesoderm, continuous with the urogenital sinus, the mesoderm will give rise to the umbilical arteries and vein
- Urachus – the remnant of the allantois that condenses and attaches the bladder to the umbilicus after birth
- Nephrogenic cord – condensation of the intermediate mesoderm close to the dorsal aorta that gives rise to the primitive kidney and associated structures
- Common nephritic duct – the portion of the mesonephric duct that attaches to the urogenital sinus after development of the ureteric bud, serves as a conduit for urine
- Gonadal ridge – a mass of mesoderm that develops due to migration of primordial germ cells from the umbilical vesicle, will form the gonads
- Cloaca – expanded portion of the caudal end of the primitive gut tube

The journey of development

- Development of the intermediate mesoderm – **Week 3**
- Folding of the embryo pinches off and forms the gut tube – **Week 4**

Developmental processes related to Urinary organs – Weeks 3 to 32

Everything happens synchronously with many simultaneous events

- Intermediate mesoderm forms urogenital ridge on either side of the dorsal aorta - **Week 3/4**
- Urorectal septum develops and starts separating the cloaca
- Formation and degeneration of pronephros
- Mesonephros and mesonephric duct develop
- Development of the ureteric bud – **Day 28**
- Ureteric bud makes contact with metanephric mesoderm and bifurcates – **Day 32**
- Fetal suprarenal cortex forms
- Mesonephric tubules fuse to mesonephric duct and become functional
- Major calyces are formed in the developing metanephros
- Neural crest cells migrate into developing suprarenal cortex – **Week 7**
- Metanephric kidney assumes its final position in the abdomen – **Week 9**
- Mesonephros stops functioning – **Week 10**

Week 4

Week 5

Week 6

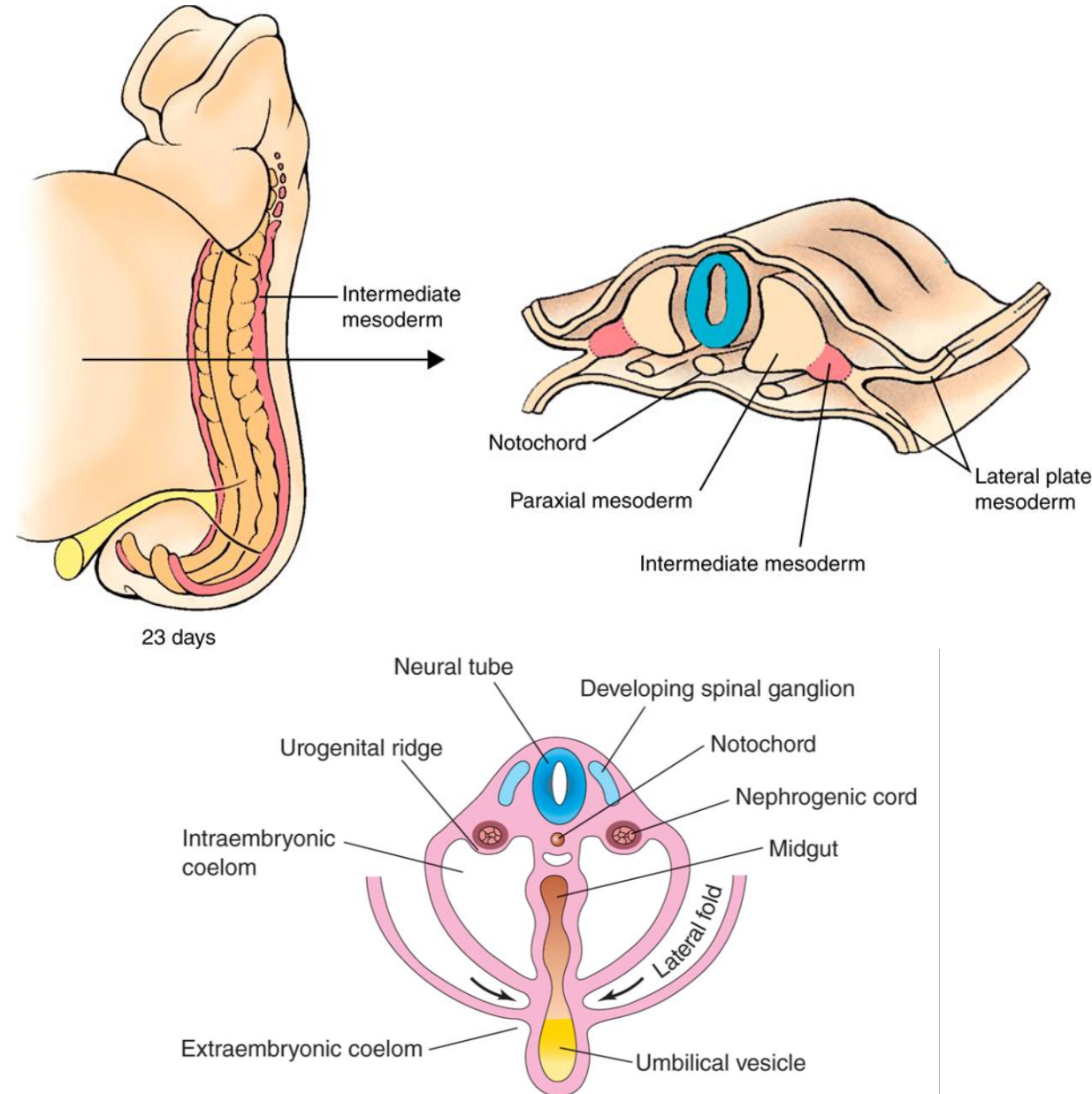


Lecture outline

- General concepts for development of the urogenital system
- Overview of the suprarenal gland
- Development of the suprarenal gland
- Overview of the gross and histological structure of the kidney
- Development of the kidneys
- Gross overview of the bladder and ureters
- Development of the bladder and ureter
- Developmental anomalies and defects of the urinary system

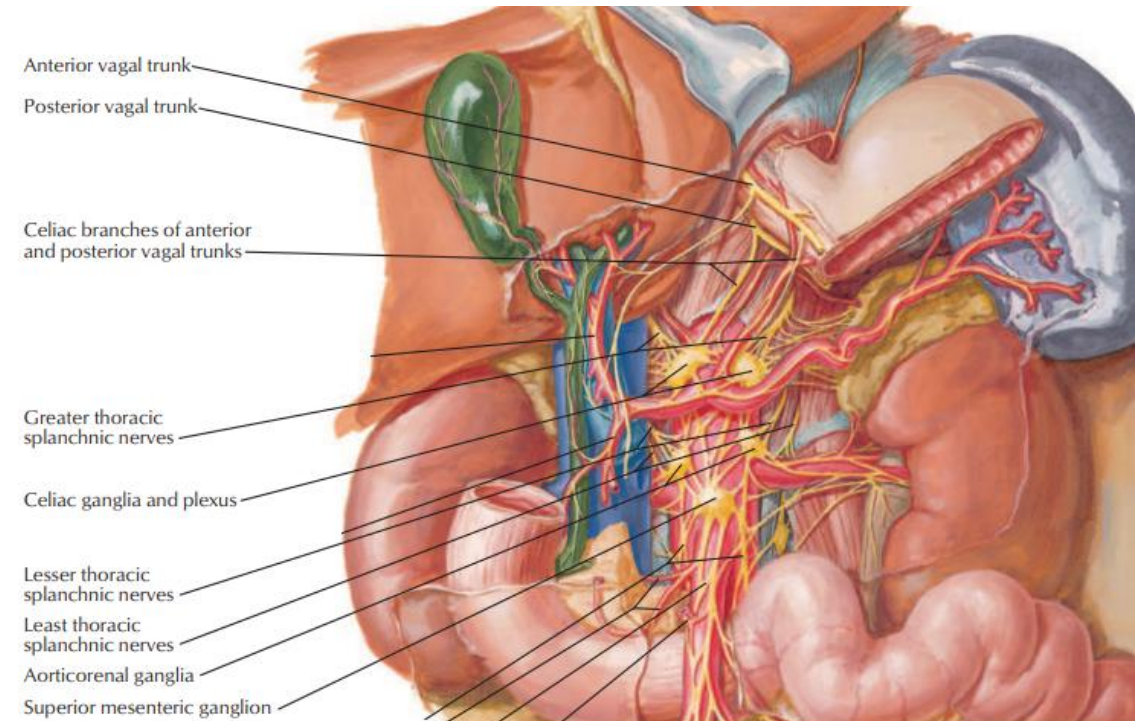
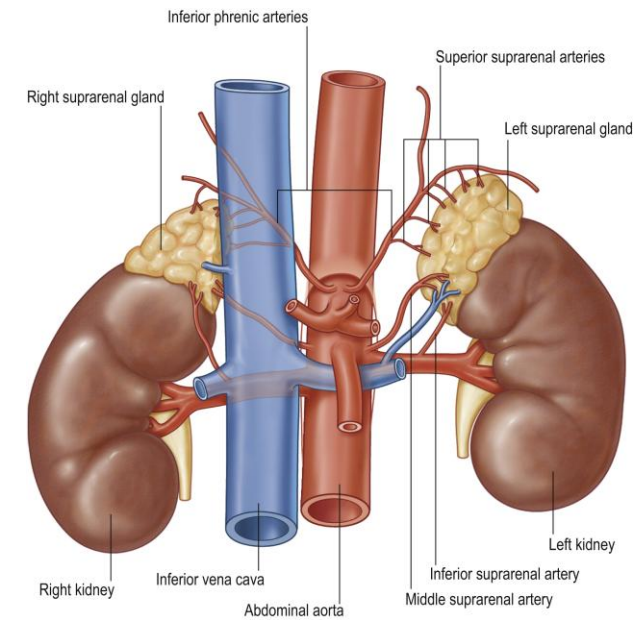
General concepts for development of the urogenital system

- Development of the urinary and reproductive system are linked
- Develops from
 - **Intermediate mesoderm**
 - Forms a ridge just lateral to the developing dorsal aorta = **urogenital ridge**
 - Deeper portion forms the **nephrogenic cord**
 - The most caudal end of the primitive gut tube
 - Expands to form **the cloaca**
 - Splanchnic mesoderm surrounding the hindgut
 - Gives rise to muscle and connective tissue of the bladder



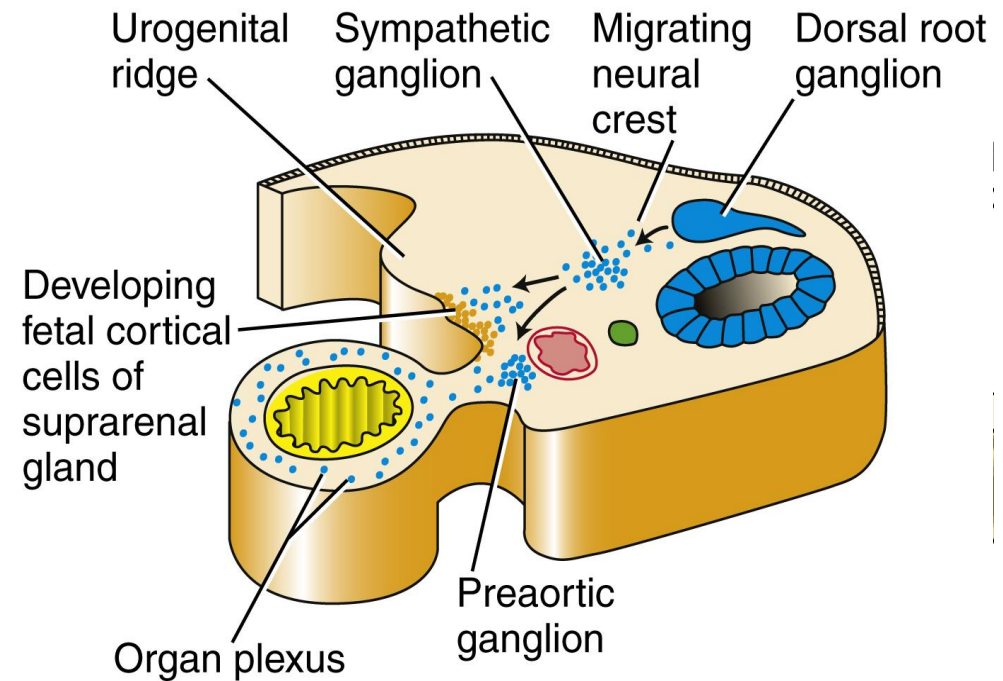
Suprarenal glands overview

- Endocrine glands located at the **superior pole of the kidney**
- The main nerve supply is by **preganglionic sympathetic fibers** from the splanchnic nerves via the celiac plexus, with fibers synapsing directly with medullary cells.
 - Blood vessels receive postganglionic sympathetic supply
 - Cortical control is not neural but by ACTH from the anterior pituitary
- Lymph drainage is to the **para-aortic nodes**
 - Receive their blood supply via branches from **Inferior phrenic arteries**
 - Directly from the **aorta**
 - **Renal artery**
- Venous drainage is via the suprarenal vein that drains
 - into the **IVC on the right**
 - into the **renal or inferior phrenic vein on the left**



Development of the suprarenal glands

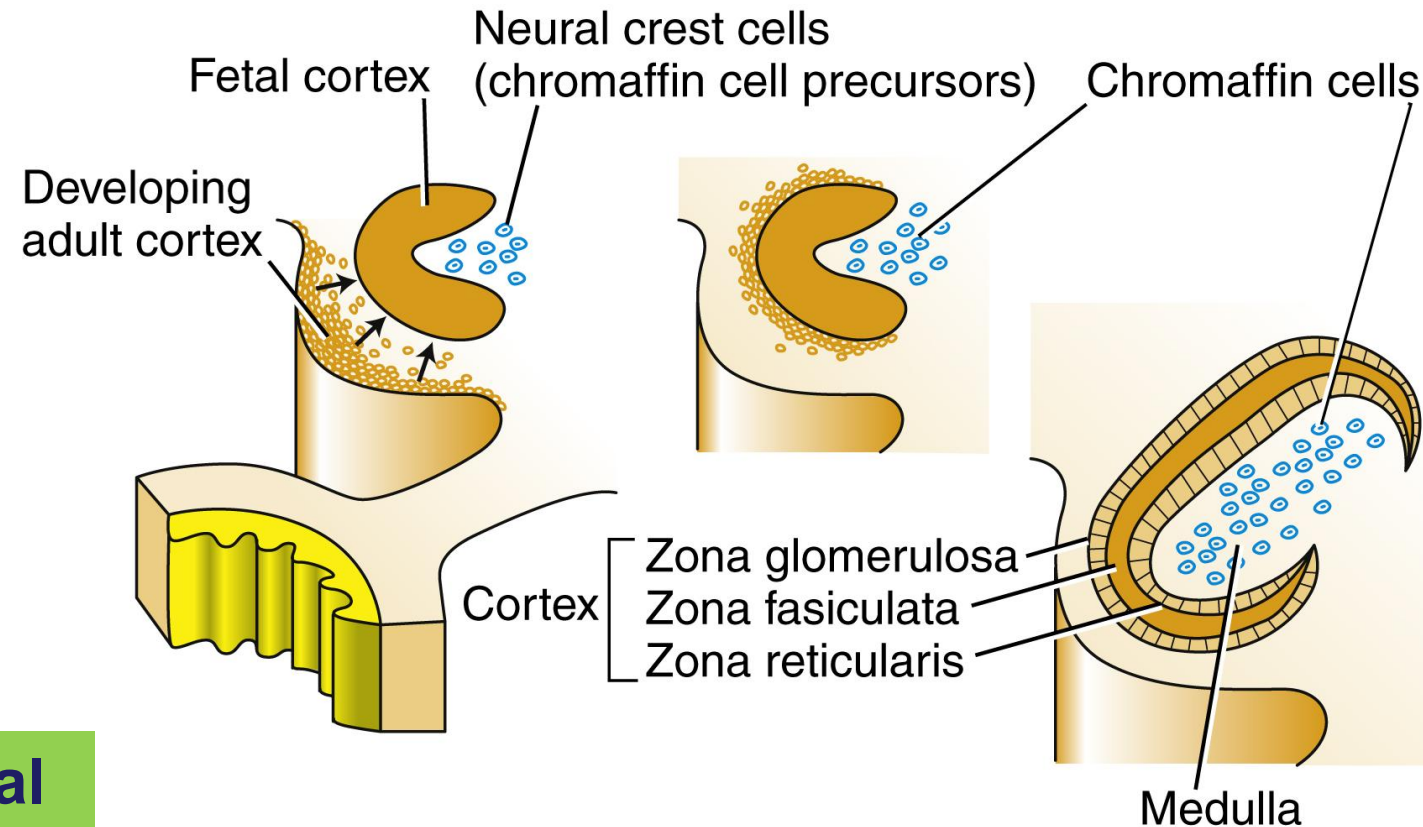
- Develops ventrally to the developing kidney
- Week 5 - **Coelomic epithelium** adjacent to the developing gonadal ridge proliferate and migrate into the underlying mesoderm
 - Differentiate into large acidophilic cells = **fetal suprarenal cortex**
 - A second set of cells develop surrounding the fetal cortex cells = **definitive cortex**
- These cells exhibit features of **steroid producing cells**
- Week 7 – **neural crest cells** migrate into the developing fetal cortex and form the medulla
- Week 8 – the suprarenal gland is fully encapsulated



Development of the suprarenal glands

- **Second trimester** = fetal cortical layer grows rapidly
 - Begins to secrete **dehydroepiandrosterone (DHEA)** converted to estrogen by the placenta (helps to maintain pregnancy)
 - Other products **influence the development of the lungs, liver and digestive tract**
- As development continues the gland decreases in size gradually

Relative to body mass the fetal suprarenal glands are 10-20x larger than the adult



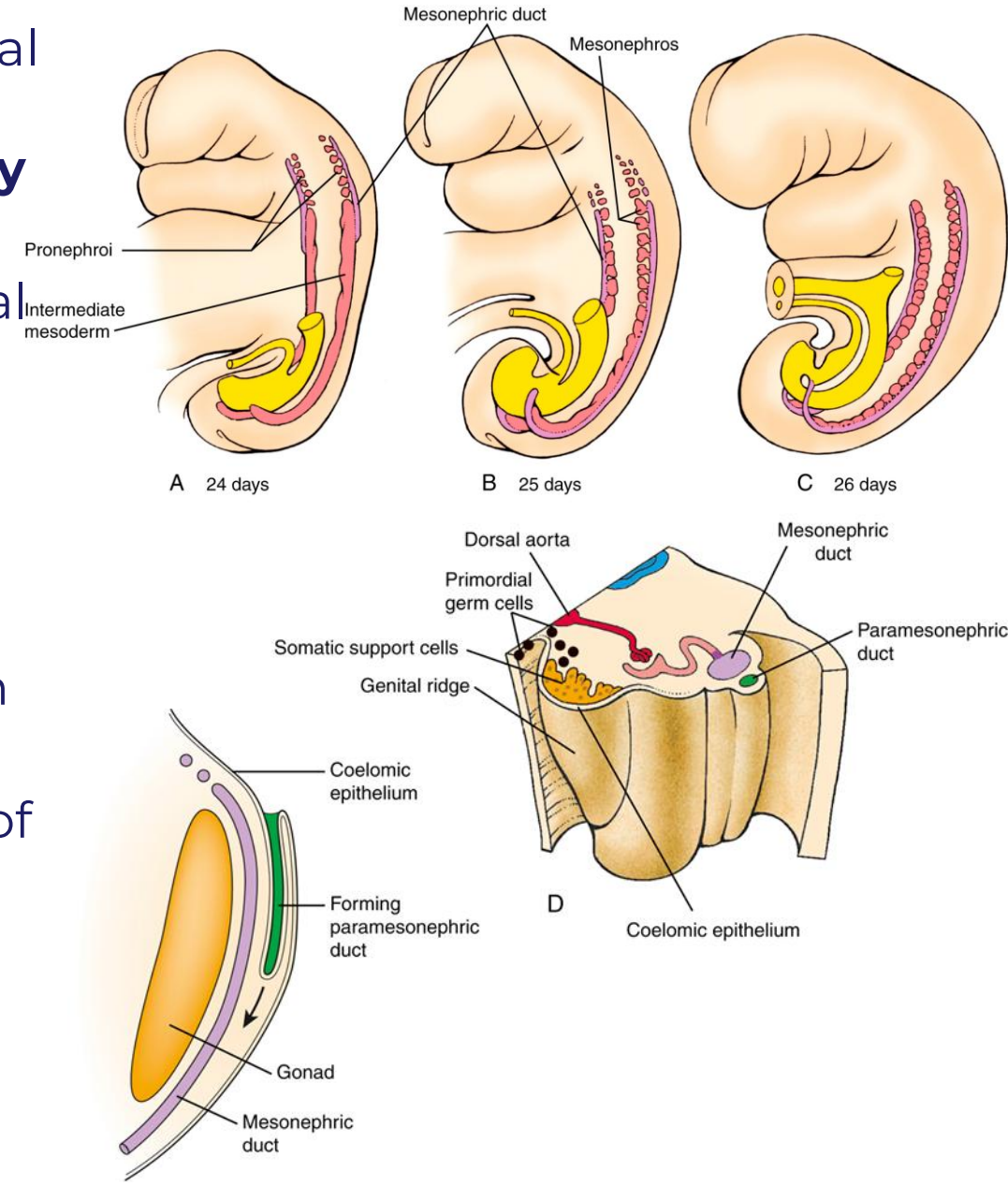
Development of the suprarenal glands at and after birth

- Cortex is divided into two portions – at birth
- The cortex regresses and organizes into 3 layers - 2 months postnatal
 - Zona glomerulosa
 - Zona fasciculata
 - Zona reticularis
- Fetal cortex completely disappears – 1 year
- Full adult gland is present – 4 years

- cortex & medulla
- minor & major calyces
- renal pelvis
- ureter

Week 4 – Mesonephric duct and pronephros

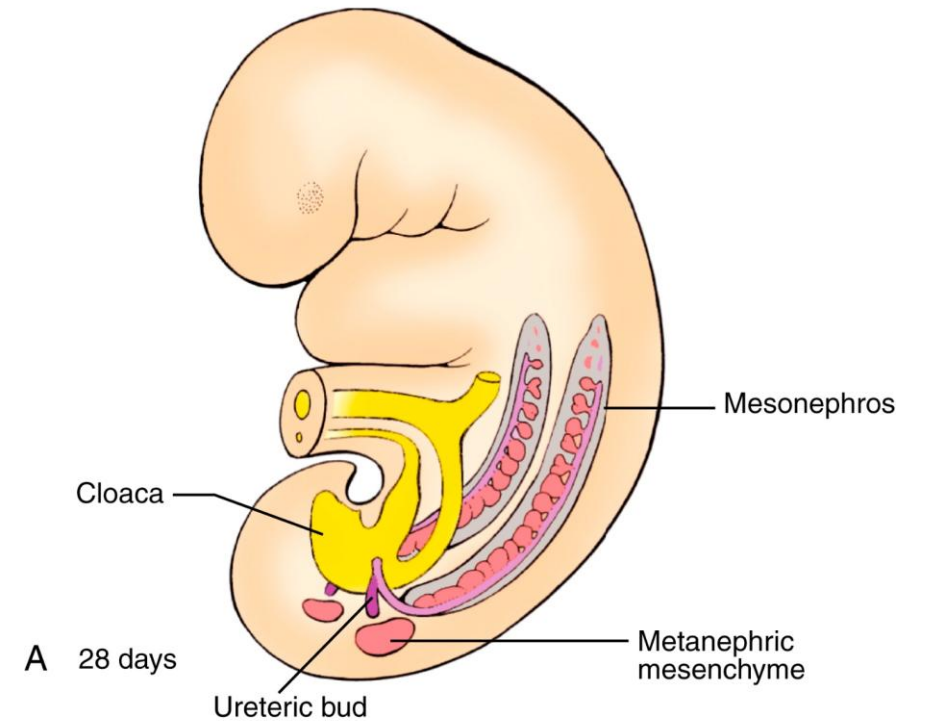
- Intermediate mesoderm at the 5th to 7th cervical levels develop into buds = **pronephros**
 - Buds become hollow and **disappear by day 24/25**
- Solid longitudinal rods** of condensed epithelial tissue form bilaterally = **mesonephric duct**
- Induces formation of **mesonephric buds**
- In the caudal most ends of the mesonephric ducts fuse with the **ventrolateral walls of the cloaca** - day 26
 - The mesonephric ducts form a **lumen** from the point of contact with the cloaca
- A **ureteric bud** develops from the caudal end of the mesonephric duct – **day 28**



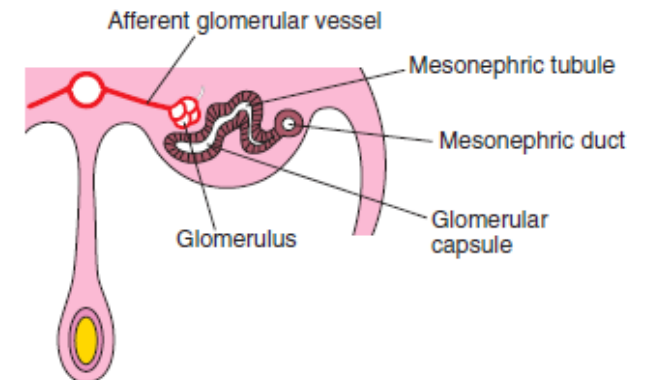
Weeks 4 – 10: Development of the primitive kidney

Mesonephros

- Approx 40 **mesonephric tubules** develop alongside the mesonephric duct
 - Craniocaudally in the thoracic and lumbar regions
 - Cranial ones regress as the caudal ones are added until end of week 5
- Week 6: 20 pairs of mesonephric tubules in the **L1-L3 region**
 - Differentiate into excretory units resembling the nephron consisting of;
 - **cup-shaped sac = Bowman's capsule**
 - **capillaries = glomerulus**
 - The lateral tips of cranial 7, fuse to the mesonephric duct – starts functioning
- Functioning kidney until **Week 10**

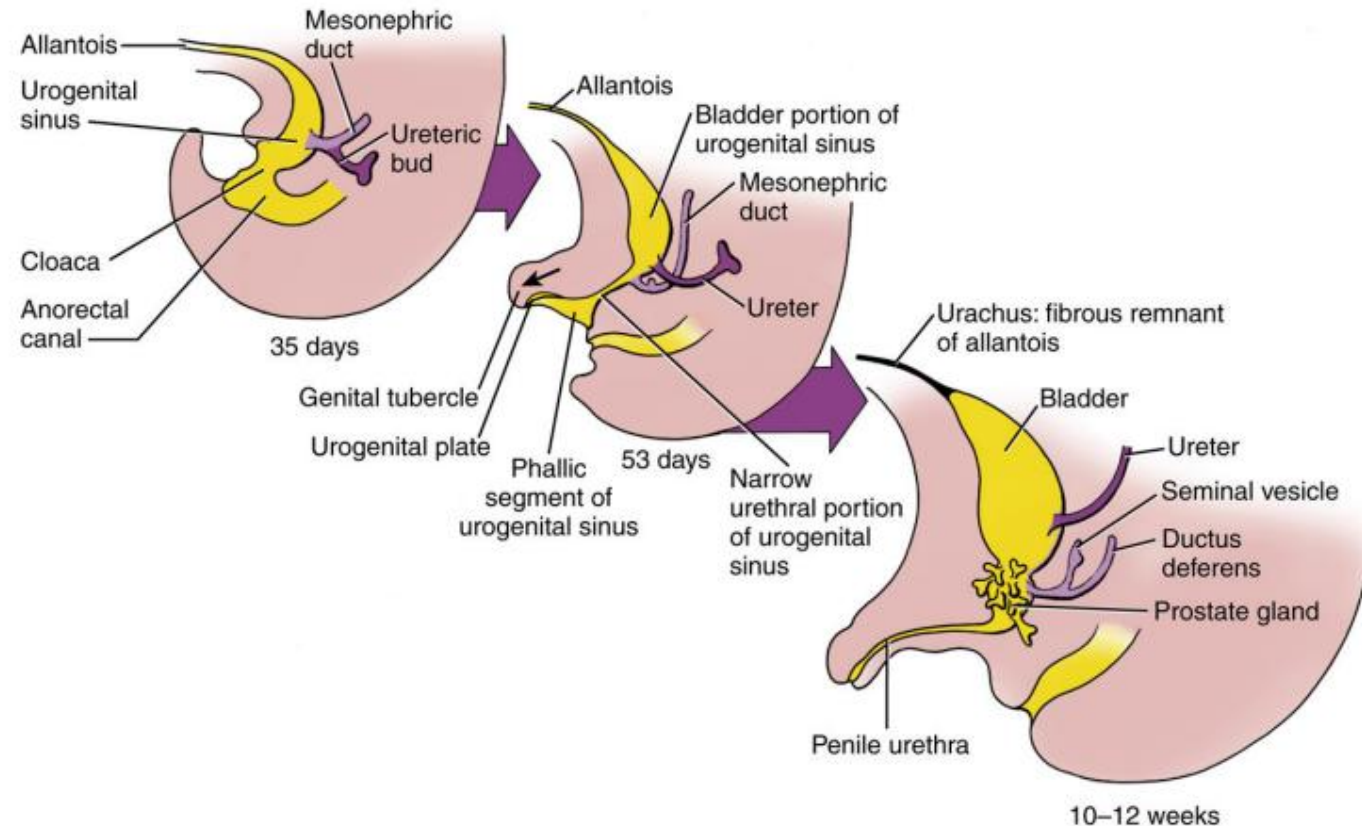


Renal corpuscle



Fate of the mesonephros

- After week 10 the mesonephric tubules regress via apoptosis
- In females the mesonephric duct also regress via apoptosis
- In males the mesonephric duct and some of the tubules persist to form important elements of the reproductive system



The metanephros – definitive kidney

- Consists of 2 functional components
- Begins with development of the ureteric bud – **Day 28**

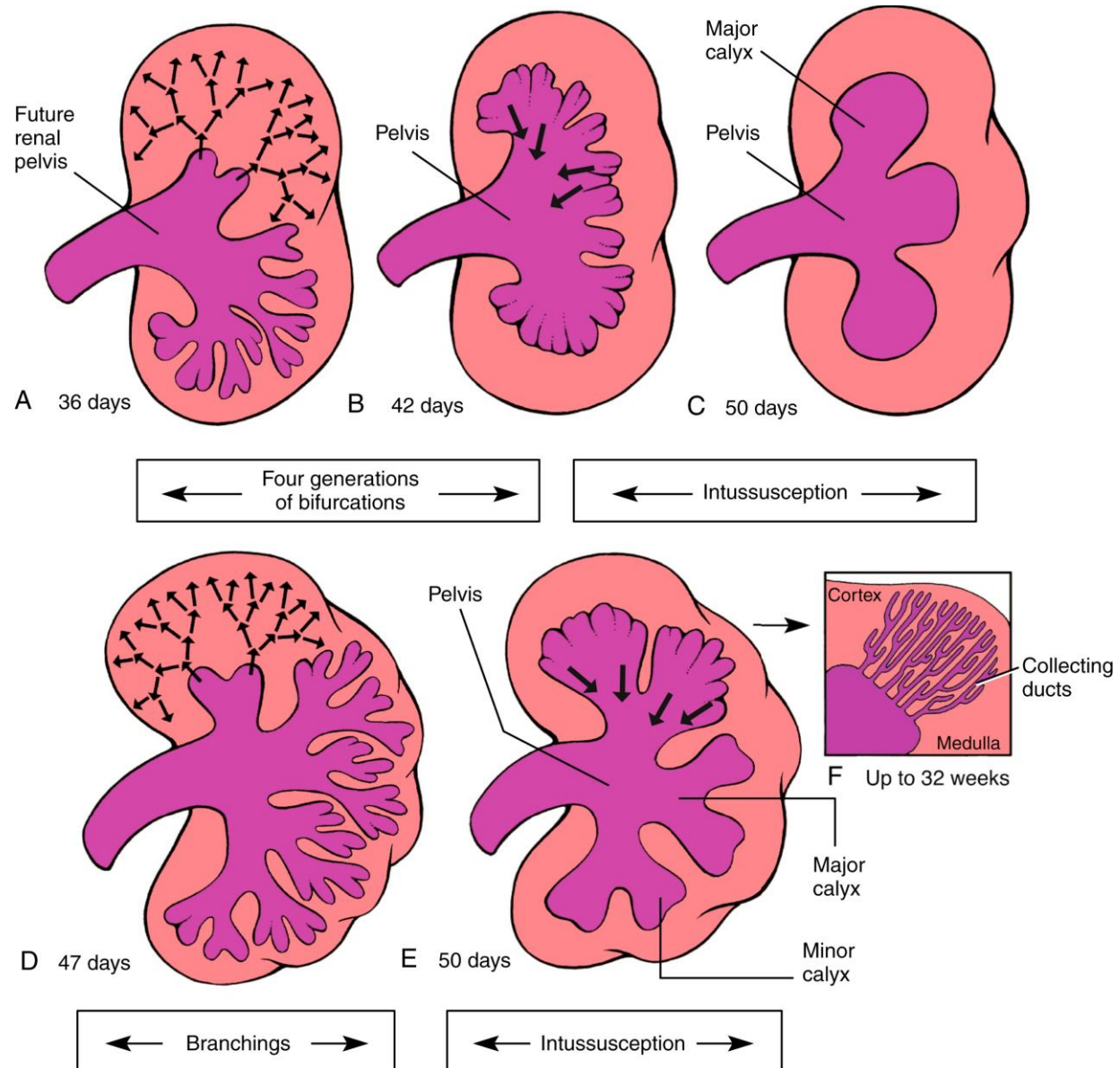
Collecting Portion	Excretory Portion (or Nephron)
Develops through successive branching of the ureteric bud	Differentiation of the intermediate mesodermal tissue in the metanephric blastema
Ureter	Bowman's capsule
Renal pelvis	Proximal convoluted tubule
Major and minor calyces	Loop of Henle
Collecting ducts	Distal convoluted tubule

Weeks 4 – 32: Development of the collecting portion of kidney

- Day 28: The **ureteric buds** grow out from the mesonephric duct
- Day 32: They grow laterally and penetrate the **metanephric mesenchyme** in the sacral region of the intermediate mesoderm
 - The buds begin to **bifurcate**
- As bifurcation occurs each ureteric tip acquires a cap-like aggregate of mesenchymal tissue the **cap mesenchyme**
- Week 6: 16 branches these fuse together to form the **major calyces**
- The next sequence of branching will fuse together to form the **minor calyces**
- Week 16 : 14-16 lobes have formed, and the fetal kidney has a lobulated appearance
- Week 32: 11 generations of branching has yielded 1-3 million branches = **collecting ducts**

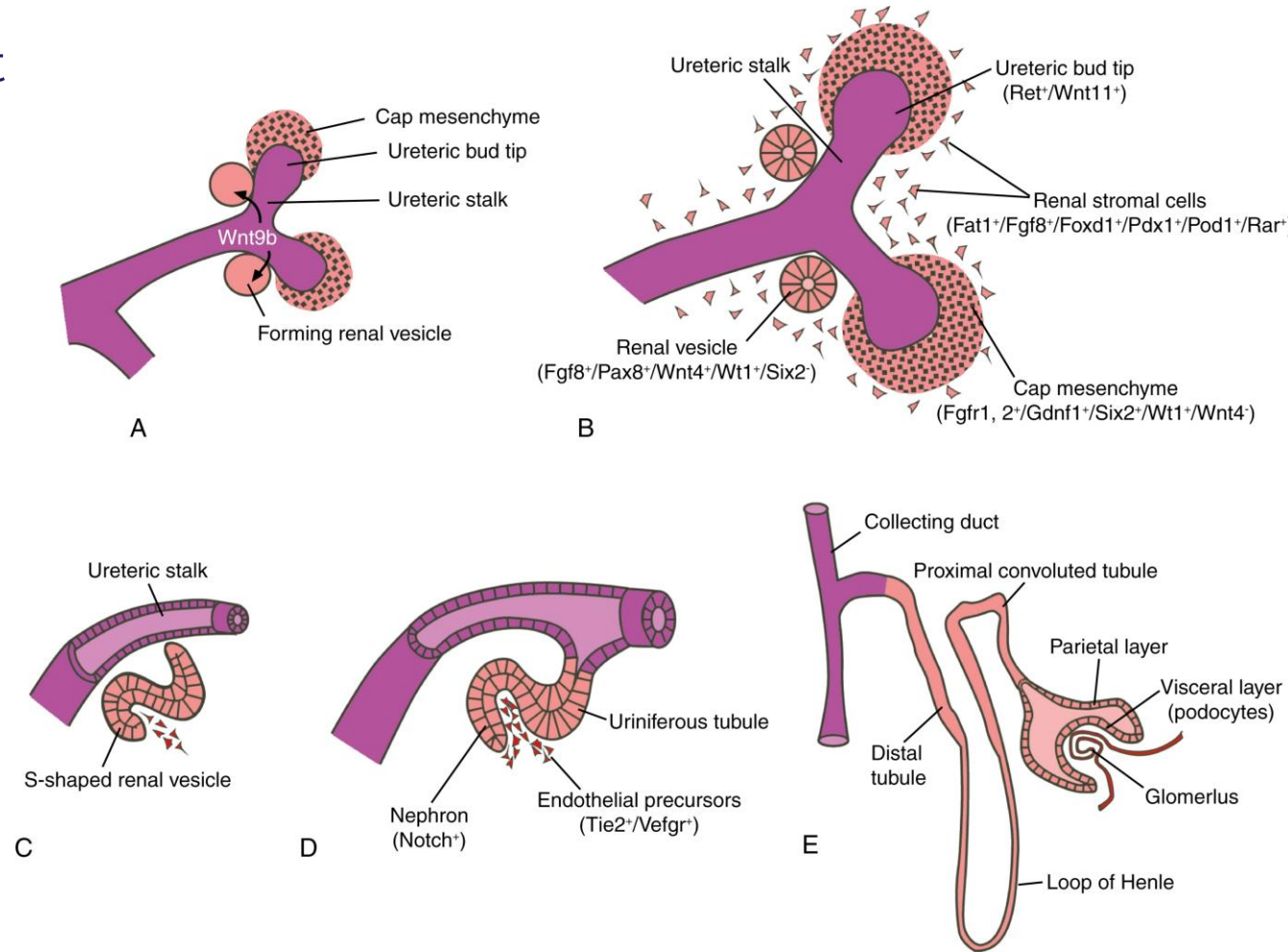
Weeks 4 – 32: Development of the collecting portion of kidney

- The **ureteric buds** grow out from the mesonephric duct
 - Forms a common **nephric duct** to the urogenital sinus
- Makes contact with the metanephric intermediate mesoderm and begins to branch
- Successive branching and selective coalescing** (intussusception) of the ureteric bud forms the **collecting system**
- Week 16: 14 – 16 lobes have formed



Development of the metanephric nephron

- Branching induces formation of the **cap mesenchyme**
 - Cells break away and lie adjacent to the ureteric stalk = **renal vesicle**
 - Modification of the renal vesicle leads to the development of the **nephron**
- The developing nephron plays a role in the orderly branching pattern seen in the ureteric bud
 - Reciprocal relationship**
- By week 10 the first fully functional nephrons appear
- Final number of nephrons are determined by the branching of the ureteric bud ~ **1-2 million**



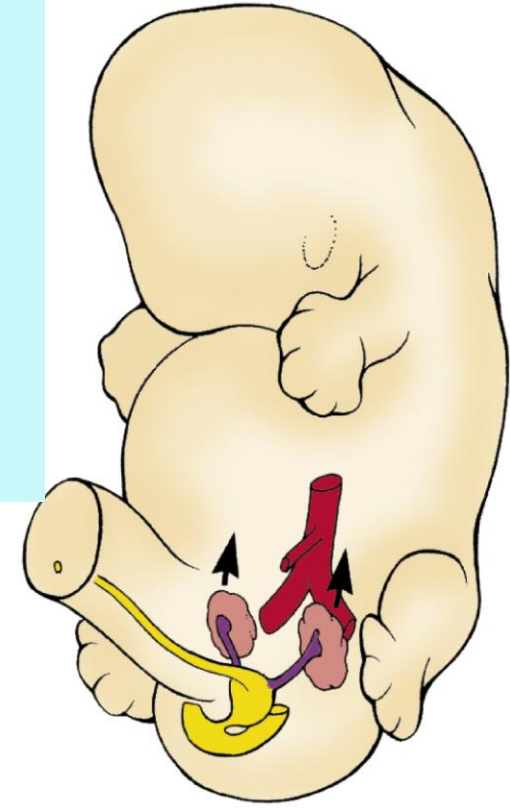
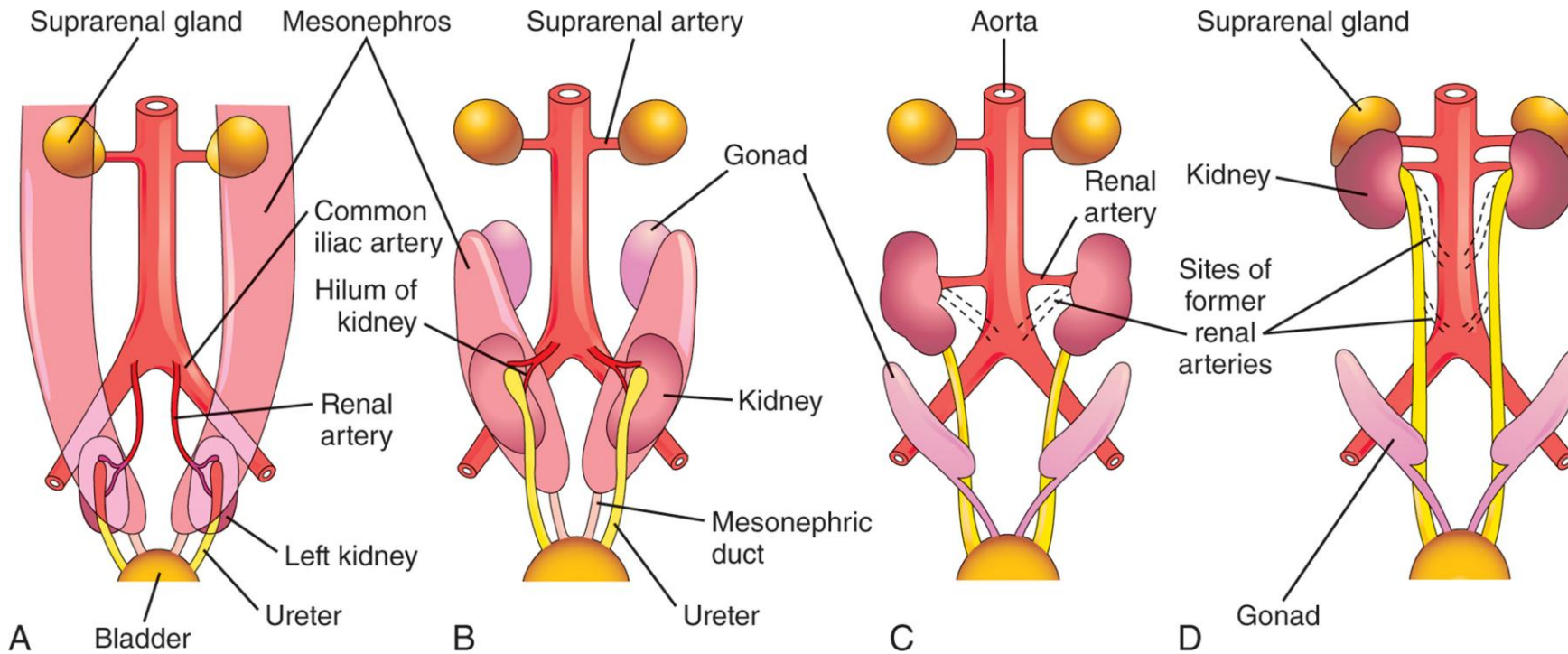
Renal vascularization and relocation “ascent”

Kidney “ascent” occurs because of the growth of the pelvic and abdominal structures

Renal arteries develop from the aorta **sequentially** as the kidneys relocate

Kidney migration ends when they reach the suprarenal glands

As they migrate, they **rotate 90°** so the hilum faces medially



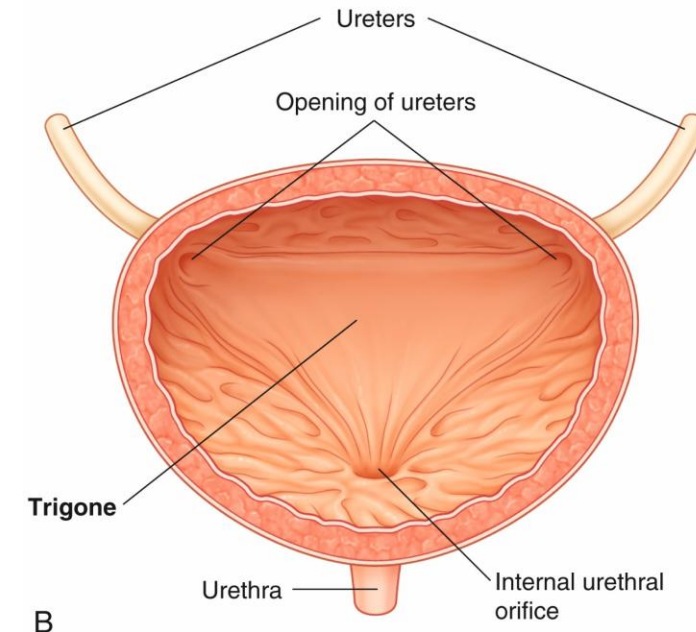
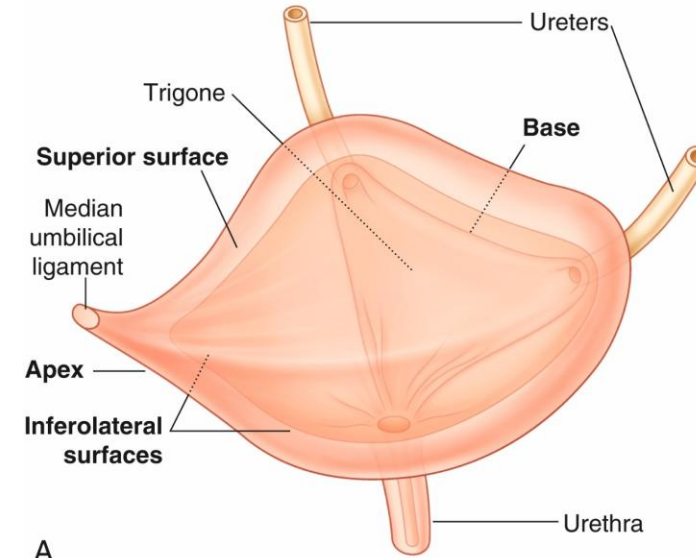
A 6th week

Renal vascularization and relocation “ascent”

- The permanent kidneys develop in the pelvis
- Growth of the abdomen and pelvis **relocates the kidneys** to their lumbar position
 - As the caudal parts of the embryo grow, the kidneys come to lie higher and higher in the abdomen (“**ascent**”)
- Initially, the hilum faces ventrally and receives branches from the common iliac arteries
 - During their upward migration, the kidneys **rotate** medially almost 90 degrees and lie lateral to the aorta
- Successively higher and higher branches from the dorsal aorta supply the kidneys as they “ascend”
- Normally the primordial caudal branches disappear and only the final renal arteries persist
- By week 9, the kidneys are in contact with the suprarenal glands and reach their adult position

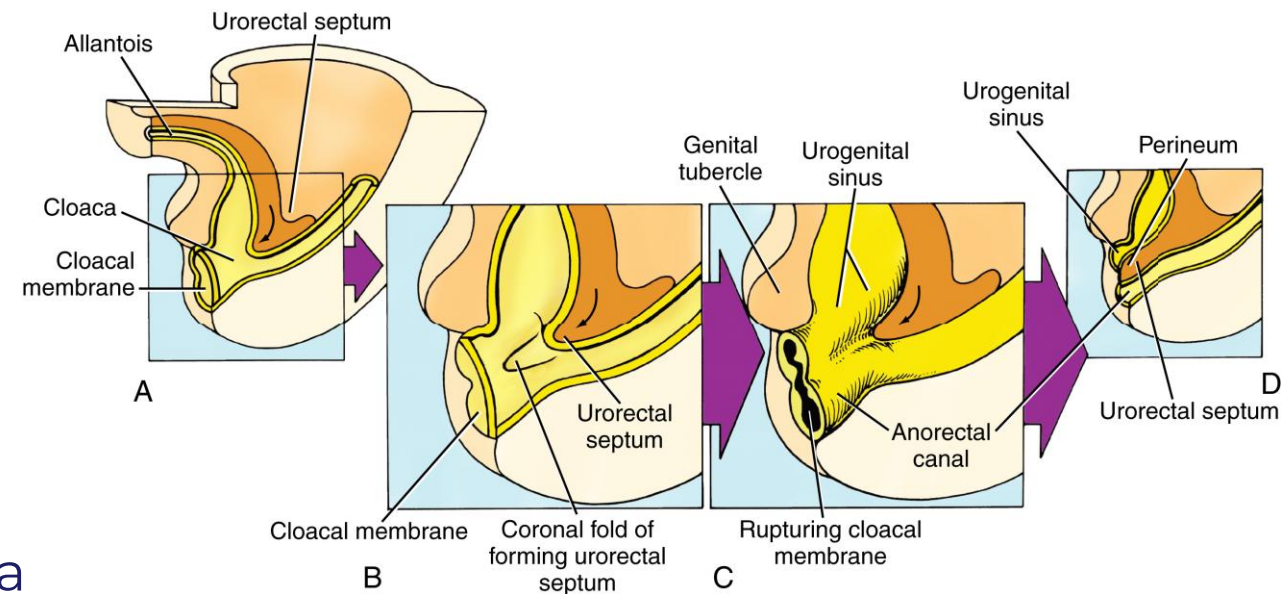
Gross Overview – Bladder and urethra

- Bladder consist of **smooth muscle** lined with **mucosa internally**
- Two ureters enter the bladder posterolaterally
- A single urethra exits inferiorly at the neck
- The area between the openings of these structure is called the **trigone**



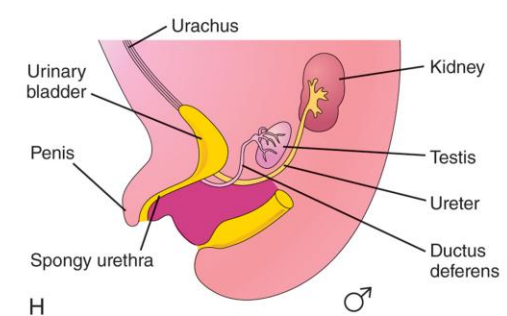
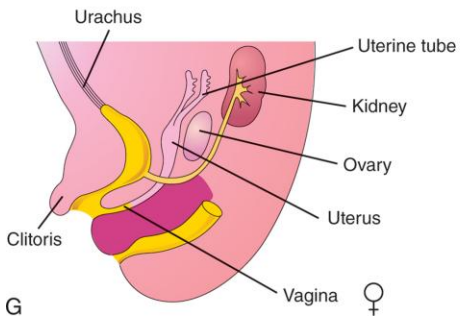
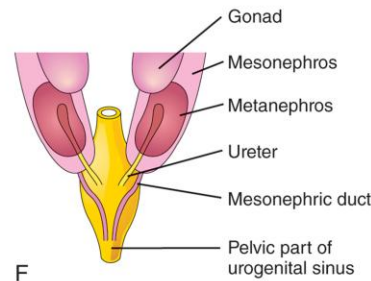
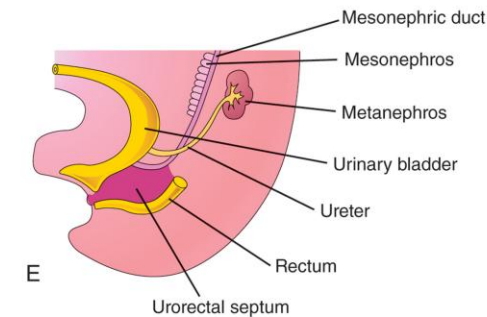
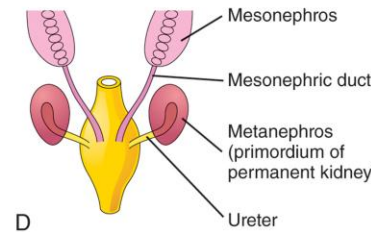
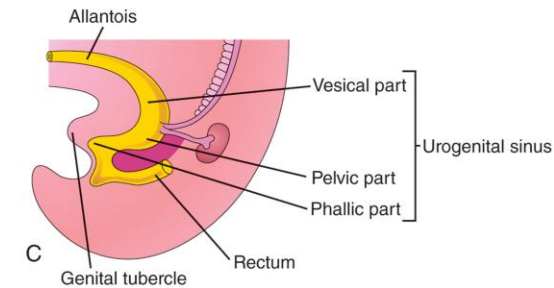
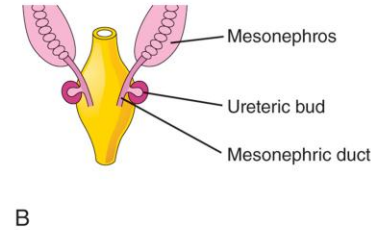
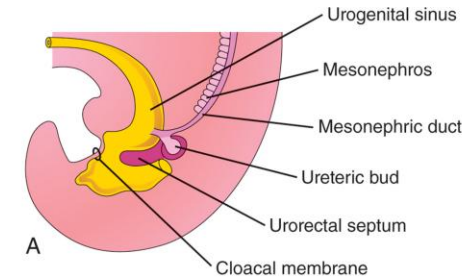
Development of the bladder and urethra

- The bladder develops from the embryonic hindgut
- Just deep to the cloacal membrane (which closes off the hindgut tube from the amniotic cavity) the hindgut expands to form the **cloaca**
- It has a diverticulum the **allantois** (which will form the umbilical blood vessels)
 - Condenses to form the **uracus = median umbilical ligament** in adults
- Weeks 4-6: the **urorectal septum**, mesoderm, develops and divides the cloaca
 - Anorectal canal (dorsal)
 - Urogenital sinus (ventral)
 - Vesicle portion
 - Forms the **bladder**
 - Narrow pelvic/urethral portion
 - **Membranous** urethra
 - **Prostatic** urethra in males
 - Phallic portion
 - **Penile urethra** in males
 - **Vestibule of vagina** in female



Development of the bladder and ureters

- The nephritic duct undergoes apoptosis
- The urogenital sinus expands, and the ureteric bud is absorbed into the bladder wall = **ureter**
 - Relocates the entry point of the ureter **superiorly and laterally**
- In **males the mesonephric duct persists**
 - Has it's own entry point at the pelvic portion of the urogenital sinus
- Week 12:** The splanchnic mesoderm that surrounds the cloaca gives rise to the muscle of the bladder



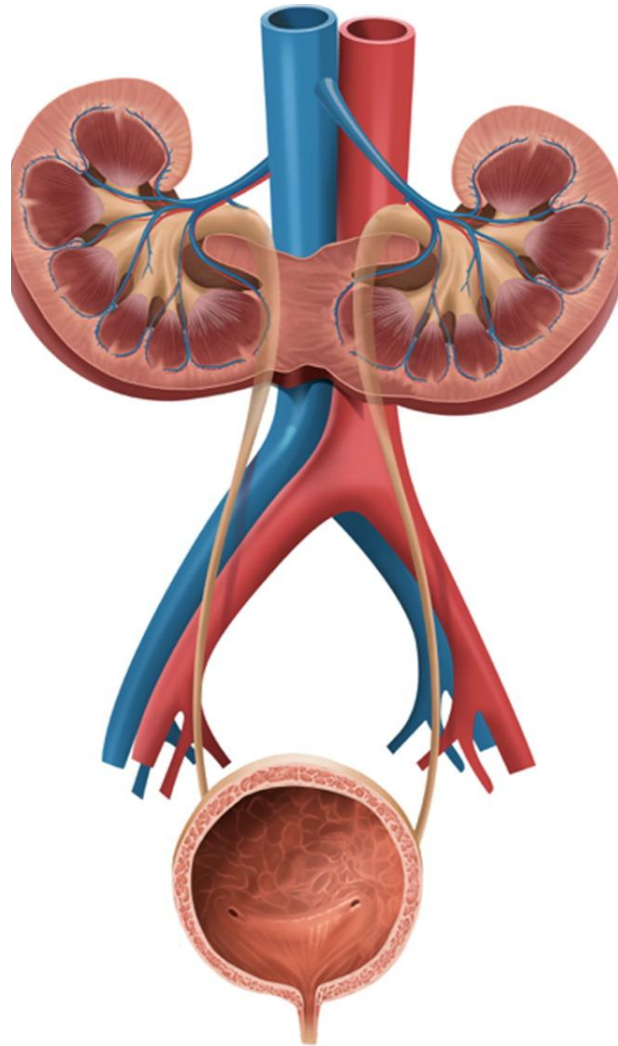
Clinical embryology

- Discoid (pancake) and horseshoe kidney
- Bifid and double ureter
- Renal agenesis
- Multiple renal arteries and their consequences

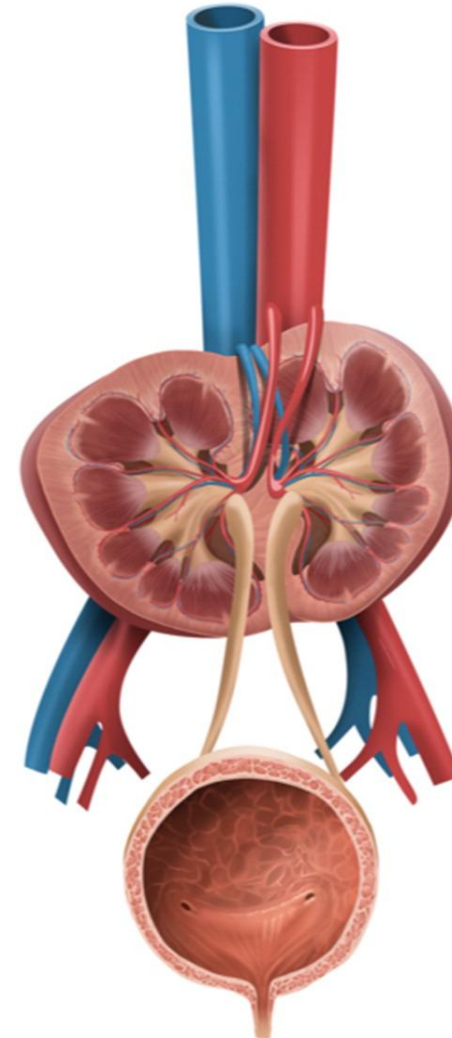
Fusion Anomalies

Horseshoe Kidney

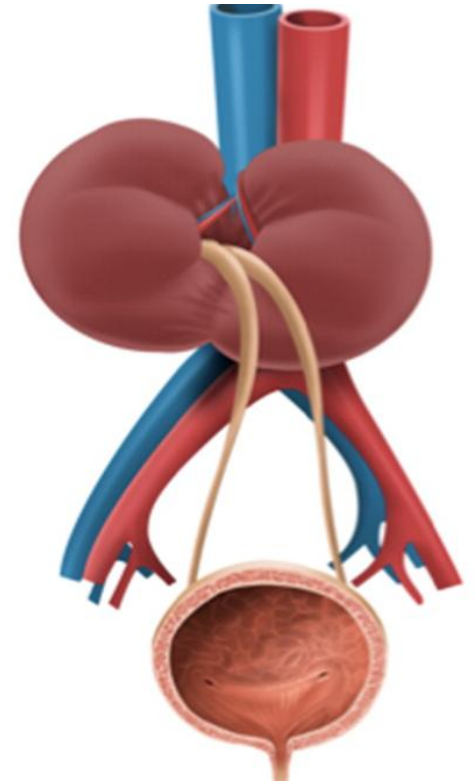
- 1 in 500 births
- Most often fusion of lower poles while still in pelvis
- ascent interrupted by the **inferior mesenteric artery**



Horseshoe kidney



Pancake kidney

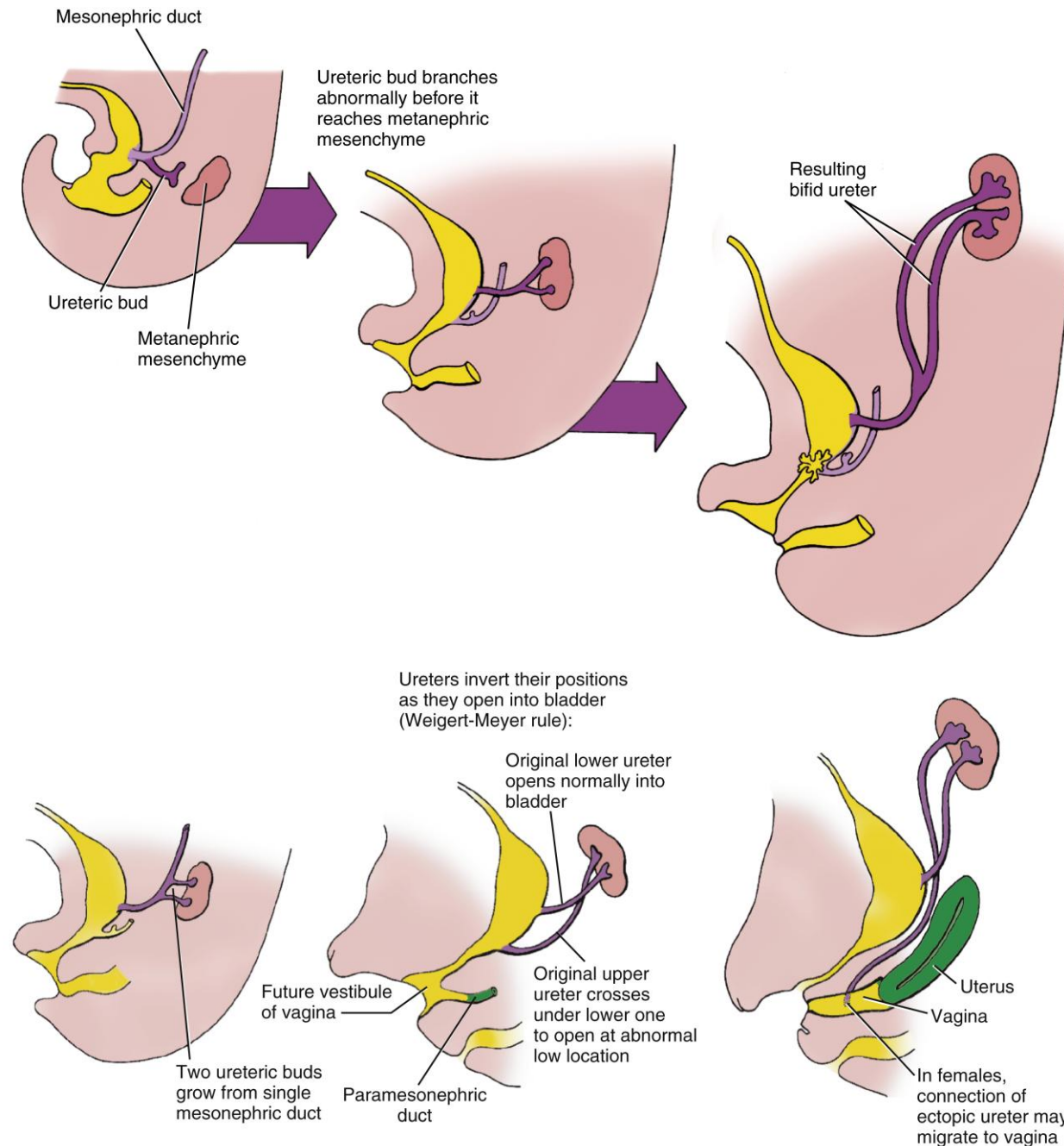


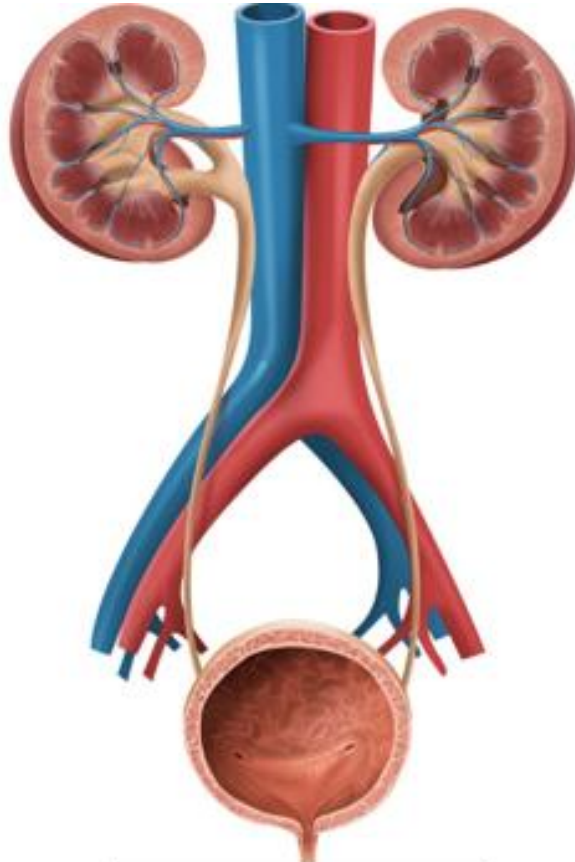
Pancake (discoid) Kidney

- 1 in 65000 births
- Fusion of superior, middle and inferior poles of both kidneys

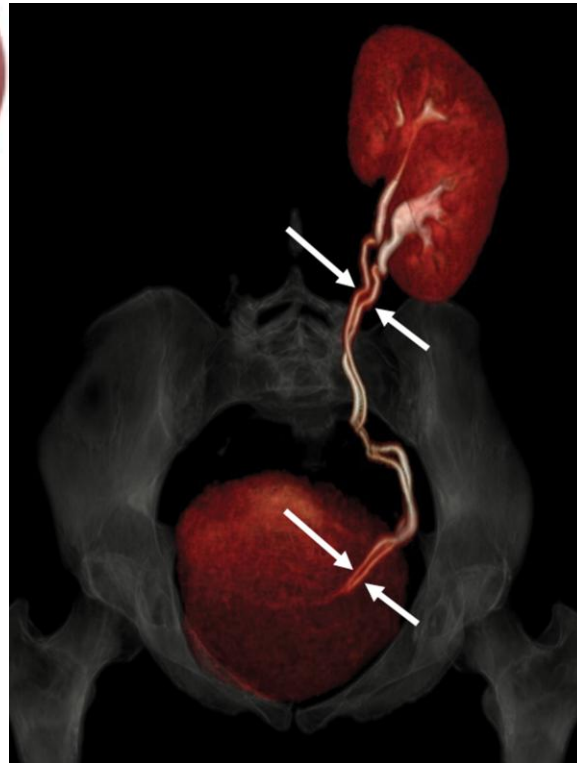
Duplicated Structures

- Bifid ureter
 - Early **incomplete division** of the ureteric bud
- Double kidney
 - Early **complete division** of the ureteric bud inducing the formation of two kidneys
- Supernumerary (extra) kidneys are rare
 - Two **separate** ureteric buds developed on that side resulting in 2 separate kidneys

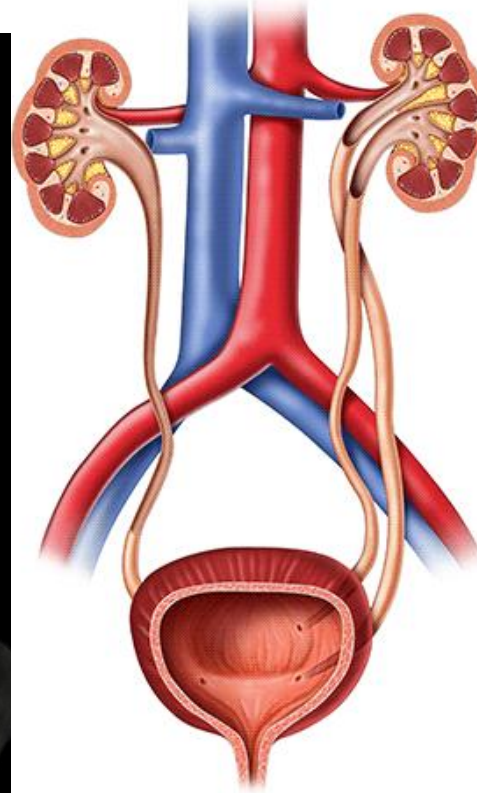




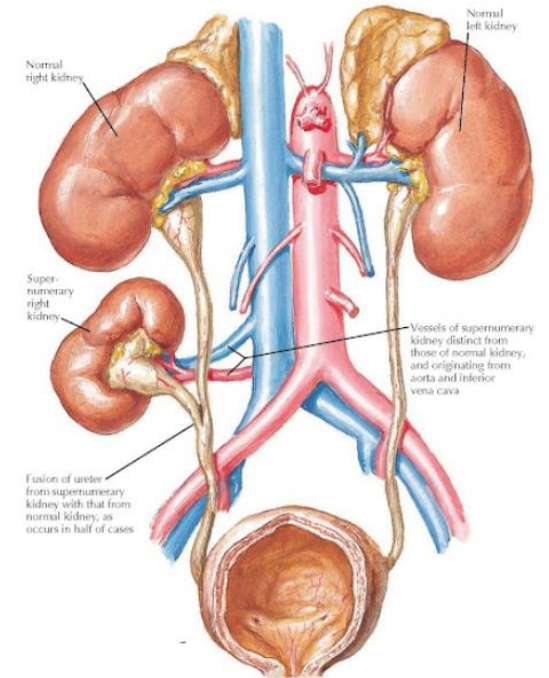
Bifid ureter:
1 collecting
system with 2
pelves



Double ureter:
2 collecting systems
with shared blood
and nerve supply



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Supernumerary
kidney: 2 collecting
systems each with
independent blood
and nerve supply

Renal Agenesis

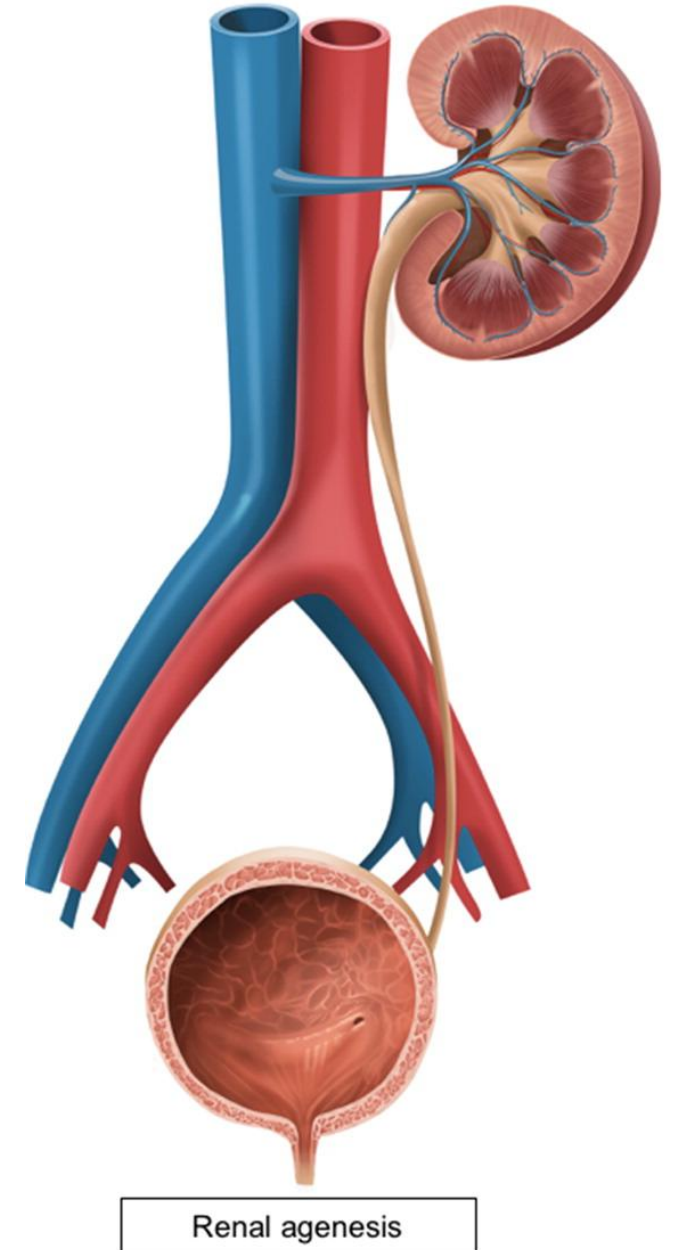
- Early **degeneration** or **failure of formation** of the ureteric bud

Unilateral:

- More common in boys
- Usually left kidney is absent
- Asymptomatic if the other kidney is normal (compensatory hypertrophy)

Bilateral:

- Can result in:
 - **Oligohydramnios**
 - Potter sequence
 - Pulmonary hypoplasia
- This is (mostly) incompatible with post-natal life

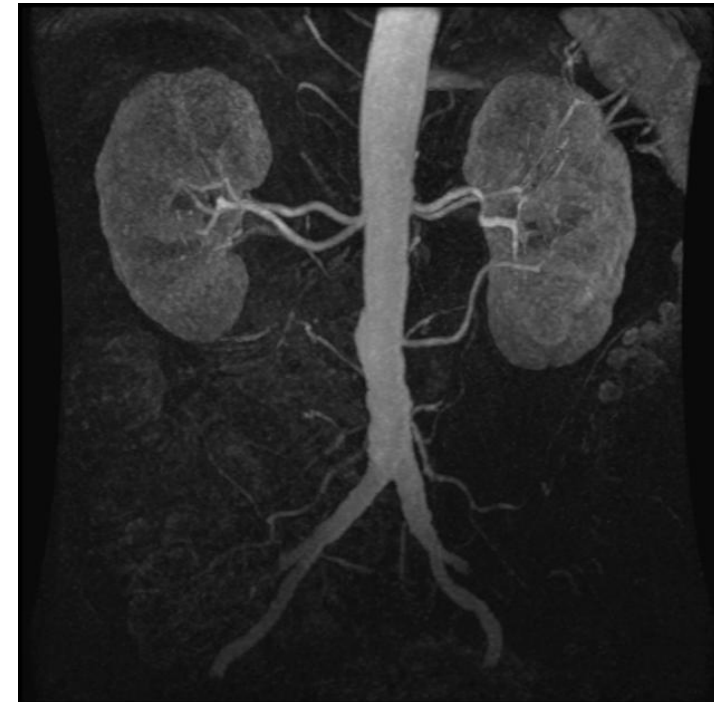
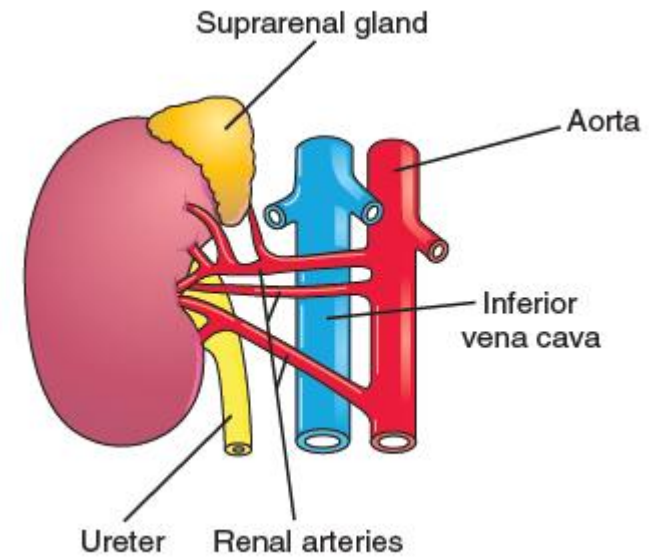


Extrarenal Arteries

Common variants of the renal arteries, typically arising from the aorta.

Subclassifications include:

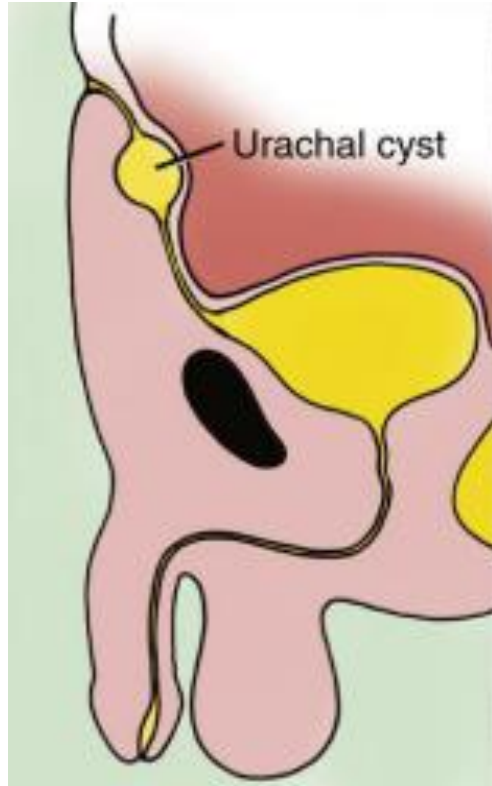
- Accessory renal artery: supplying the renal hilum
 - present in 25% of the population
 - present bilaterally in 10%
- Aberrant renal artery: supplying the superior and/or inferior pole of the kidney



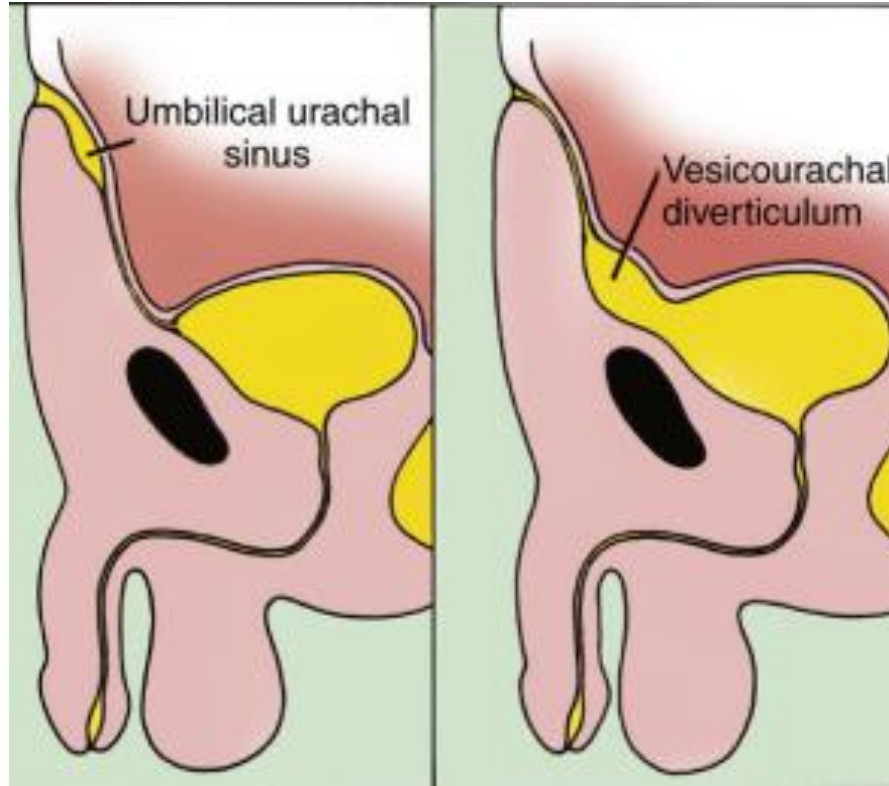
- Aberrant renal arteries at the inferior pole cross anteriorly to the ureter
 - may cause **hydronephrosis** due to physical obstruction of urine flow
- Symptoms may include colicky mid-abdominal pain, nausea and vomiting
- Also, the patient may have significant weight loss and a **palpable, ptotic kidney**



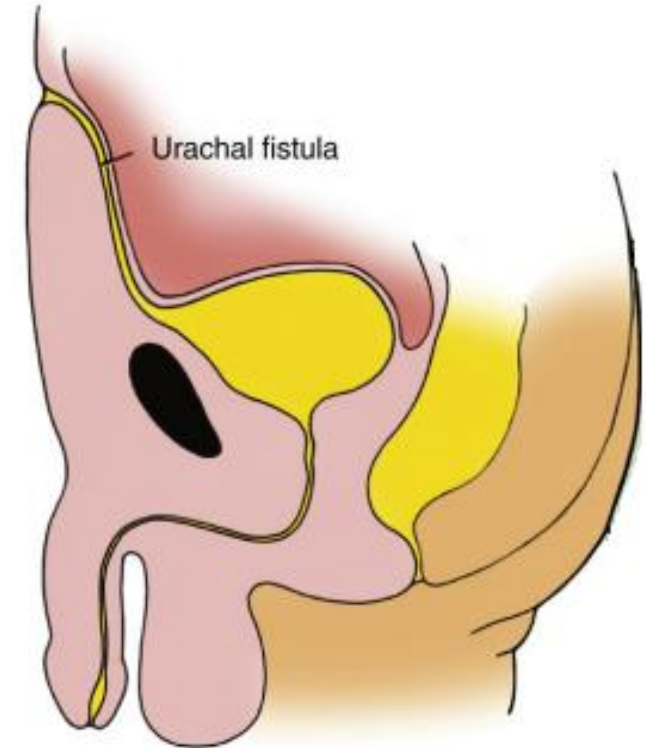
Congenital Urachal Anomalies



Urachal cyst: ends of urachal lumen close but a portion remains patent and fluid-filled



Sinus or Diverticulum: non-communicating dilatation at one end



Urachal fistula: open channel between bladder and umbilicus